

2022

DELHI Term-II

CBSE Solved Paper

Time allowed : 2 hours

Maximum Marks : 40

GENERAL INSTRUCTIONS:

Read the following instructions carefully and strictly follow them:

- This question paper contains **15** questions. **All** questions are compulsory.
- This question paper is divided into **three** Sections viz. Section **A**, **B** and **C**.
- Section **A** — Question numbers **1** to **7** are short answer type questions. Each question carries **two** marks.
- Section **B** — Question numbers **8** to **13** are also short answer type questions. Each question carries **three** marks.
- Section **C** — Question numbers **14** and **15** are case based questions. Each question carries **four** marks.
- Internal choices have been provided in some questions. Only one of the alternatives has to be attempted.

SECTION-A

- “Carbon prefers to share its valence electrons with other atoms of carbon or with atoms of other elements rather than gaining or losing the valence electrons in order to attain noble gas configuration.” Give reasons to justify this statement. **(2 Marks)**
- The atomic number of an element ‘X’ is 11.
 - Write the electronic configurations of X and find its valency.
 - Write the formula and nature of its oxide.**[OS]* (2 Marks)**
- Give reasons:
 - Placenta is extremely essential for foetal development.
 - Uterine lining becomes thick and spongy after fertilisation.**(2 Marks)**
- Name the reproductive and non-reproductive parts of bread mould (*Rhizopus*).
 - List any two advantages of vegetative propagation.**(2 Marks)**
- Name the reproductive parts of an angiosperm. Where are these parts located? Explain the structure of its male reproductive part. **(2 Marks)**

OR

What is puberty? Mention any two changes that are common to both boys and girls in early teenage years.

6. (a) Name the poles P, Q, R and S of the magnets in the following figures ‘a’ and ‘b’: **(2 Marks)**

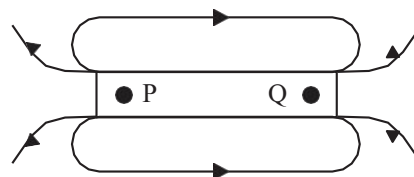


Figure a

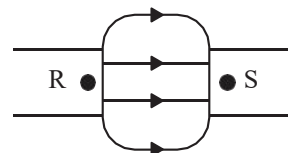


Figure b

- (b) State the inference drawn about the direction of the magnetic field lines on the basis of these diagrams.

OR

When is the force experienced by a current — carrying straight conductor placed in a uniform magnetic field.

- Maximum;
- Minimum.

7. In the following food chain, only 2J of energy was available to the peacocks. How much energy would have been present in Grass? Justify your answer. **(2 Marks)**

Grass → Grasshopper → Frog → Snake → Peacock

*[OS] denotes Out of the Syllabus questions.

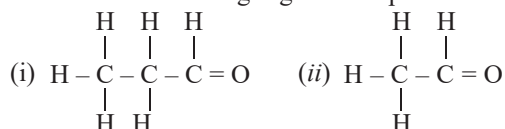
OR

- (a) What is meant by garbage? List two classes into which garbage is classified.
- (b) What do we actually mean when we say that the “enzymes are specific in their action”?

SECTION-B

8. (a) State Newland Law of Octaves.
(b) With an example, explain Dobereiner's Triads.
(c) List one limitation each of both the attempts mentioned in 'a' & 'b'. [OS]* (1 + 1 + ½ + ½ = 3 Marks)

9. Consider the following organic compounds: (3 Marks)



- (a) Name the functional group present in their compounds.
(b) Write the general formula for the compounds of this functional group.
(c) State the relationship between these compounds and draw the structure of any other compound having similar functional group.

OR

- (a) Draw the electron dot structure for ethyne.
(b) List two differences between the properties exhibited by covalent compounds and ionic compounds.
10. (a) Name the two types of gametes produced by men. (½ + ½ Mark)
(b) Does a male child inherit X chromosome from his father? Justify. (½ + 1 Marks)
(c) How many types of gametes are produced by a human female? (½ Mark)
11. (a) State Ohm's Law. Represent it mathematically.
(b) Define 1 ohm.
(c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends? (3 Marks)
12. (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends.
(b) The resistance of a wire of 0.01 cm radius is 10 Ω. If the resistivity of the wire is $50 \times 10^{-8} \Omega \text{ m}$, find the length of this wire. (2 + 1 Marks)

OR

- (a) What is the meaning of electric power of an electrical device? Write its SI unit.
(b) An electric kettle of 2kW is used for 2h. Calculate the energy consumed in
(i) kilowatt hour and (ii) joules.
13. (a) We do not clean ponds or lakes, but an aquarium needs to be cleaned regularly. Why?

- (b) Why is the ozone layer getting depleted at the higher levels of the atmosphere? Mention one harmful effect caused by its depletion. (1 + 2 Marks)

SECTION-C

This section has 02 case based questions (14 and 15).

Each case is followed by 03 sub-questions (a, b and c).

Part (a) and (b) are compulsory. However an internal choice has been provided in Part (c).

14. Mendel blended his knowledge of Science and mathematics to keep the count of the individuals exhibiting a particular trait in each generation. He observed a number of contrasting visible characters controlled in pea plants in a field. He conducted many experiments to arrive at the laws of inheritance.
- (a) What do the F_1 progeny of tall plants with round seeds and short plants with wrinkled seeds look like?
- (b) Name the recessive traits in above case.
- (c) Mention the type of the new combinations of plants obtained in F_2 progeny along with their ratio, if F_1 progeny was allowed to self pollinate.

(1 + 1 + 2 = 4 Marks)

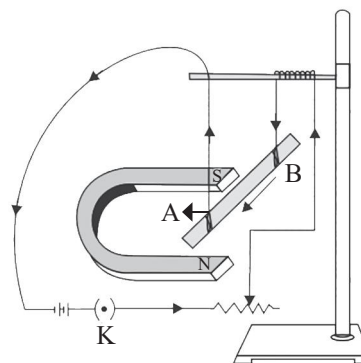
Or

If 1600 plants were obtained in F_2 progeny, write the number of plants having traits:

- (i) Tall with round seeds
(ii) Short with wrinkled seeds

Write the conclusion of the above experiment.

15. A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions: (4 Marks)



*[OS] denotes Out of the Syllabus questions.

- (a) Why does the rod get displaced on passing current through it?
- (b) State the rule that determines the direction of the force on the conductor AB.
- (c) (i) If the U shaped magnet is held vertically and the aluminum rod is suspended horizontally with its end B towards due north, then on passing current through the rod from B to A as shown, in which direction will the rod be displaced?

- (ii) Name any two devices that use current carrying conductors and magnetic field.

Or

Draw the pattern of magnetic field lines produced around a current carrying straight conductor held vertically on a horizontal cardboard. Indicate the direction of the field lines as well as the direction of current flowing through the conductor.

1) Carbon has four electrons in its outermost shell and it needs to gain or lose four electrons to attain stable noble gas configuration. Both gaining and losing four electrons is difficult for carbon because gaining will make it difficult for nucleus with six protons to hold on to 10 electrons and losing will require large amount of energy. **(1 Mark)**

Thus, carbon overcomes this problem by sharing its valence electrons with other atoms of carbon or with atoms of other elements. **(1 Mark)**

2) (i) Atomic Number of Element 'X' = 11

- The electronic configuration of 'X' is **2, 8, 1** (2 electrons in the first shell, 8 in the second, and 1 in the third).
- Valency: The number of electrons in the outermost shell is **1**, so the valency of 'X' is **1**.

(ii) **Formula and Nature of Oxide:**

- Element 'X' forms an oxide with oxygen. Since its valency is 1 and oxygen has a valency of 2, the formula of the oxide is **X₂O**.
- The oxide is **basic in nature**, as it is formed by a metal (sodium, in this case).

So, element 'X' is sodium (Na). Its oxide is sodium oxide (**Na₂O**), which is a basic oxide.

3) (i) Reasons for placenta being essential for foetal development:

- a) Placenta helps in the exchange of oxygen and nutrients between the mother and the developing foetus. **(½ Mark)**
- b) Placenta removes waste products from the foetal bloodstream. **(½ Mark)**

Without a functional placenta, the foetus would not be able to survive and develop properly.

(ii) Reasons for the uterine lining becoming thick and spongy after fertilisation:

- The thickened uterine lining provides a suitable environment for the fertilised egg to implant and grow. **(½ Mark)**
- The thickened uterine lining helps to protect the embryo from mechanical damage and infection. **(½ Mark)**

4) (a) • Reproductive parts: Sporangiothecia and sporangia. **(½ Mark)**

a) Non-reproductive parts: Mycelium and rhizoids **(½ Mark)**

(b) Advantages of vegetative propagation are:

- It allows for the production of genetically identical offspring, which can be advantageous in agriculture and horticulture for maintaining desirable traits. **(½ Mark)**

- It can be a faster and more efficient method of plant propagation compared to sexual reproduction, as it does not require the development of flowers, fruits, and seeds. **(½ Mark)**

5) Reproductive parts of an angiosperm:

- a) Male reproductive part: Stamen (consisting of anther and filament).
 - b) Female reproductive part: Carpel (consisting of stigma, style, and ovary) **(½ Mark)**
- These reproductive parts are located in the flower of an angiosperm. **(½ Mark)**

Structure of male reproductive part (stamen):

- c) Filament: It is a long, slender stalk that supports the anther.
- d) Anther: It is a sac-like structure located at the tip of the filament that contains the pollen grains. **(1 Mark)**

OR

- e) Puberty is the process of physical maturation where an adolescent reaches sexual maturity and become capable of reproduction.

It is a period of rapid growth and physical changes that typically occurs between the ages of 8 and 13 in girls, and 9 and 14 in boys. **(1 Mark)**

Two changes that are common to both boys and girls during early teenage years are:

- i) Both boys and girls experience a rapid growth spurt during puberty. **(½ Mark)**
- ii) Growth of hairs in some parts of the body such as armpits and genital areas. **(½ Mark)**

6) (a) Figure a: P – North pole, Q – South pole **(½ Mark)**
Figure b: R – North pole, S – South pole **(½ Mark)**

- (b) The magnetic field lines emerge from north pole and merge at the south pole. **(1 Mark)**

OR

The force experienced by a current-carrying straight conductor placed in a uniform magnetic field depends on the angle between the direction of the current and the direction of the magnetic field.

- (i) The force experienced by a current-carrying straight conductor is maximum when the direction of the current is perpendicular to the direction of the magnetic field. **(1 Mark)**
 - (ii) The force experienced by a current-carrying straight conductor is minimum when the direction of the current is parallel to the direction of the magnetic field. **(1 Mark)**
- 7) • According to the 10% law of energy transfer, only 10% of the energy is transferred from one trophic level to the next.
- a) Energy available to peacock = 2J
 - b) Energy available to snake (next trophic level) = 10 times more than peacock = 20J
 - c) Energy available to frog = 10 times more than snake = 200J
 - d) Energy available to grasshopper = 10 times more than frog = 2000J
 - e) Energy available to grass (producer) = 10 times more than grasshopper = 20,000J
- Therefore, the energy present in the grass in the given food chain would be 20,000 J. **(2 Marks)**

OR

- (a) Garbage is defined as the waste material that is generated from human activities and is no longer useful. It can include a variety of materials such as food scraps, paper, plastic, glass, metal, and other items that people discard.

Garbage can be classified into two classes based on their ability to decompose:

1. Biodegradable garbage: This type of garbage can be decomposed naturally by microorganisms. E.g., food waste, paper.
 2. Non-biodegradable garbage: This type of garbage cannot be decomposed naturally. E.g., plastic, glass. **(1 Mark)**
- (b) Enzymes are specific in their action, we mean that each enzyme can only catalyse a specific chemical reaction, and will only interact with its specific substrate.

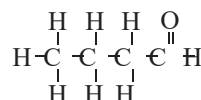
This specificity is important because specific enzymes are needed for the breakdown of a particular substance. For example, we cannot get any energy if we try to eat coal because there are no enzymes in our digestive system that can break down coal. **(1 Mark)**

8. Out of the Syllabus

9. (a) Aldehydic functional group is present in both structures. **(½ Mark)**
- (b) General formula of the aldehydic functional group: $C_nH_{2n}O$. **(½ Mark)**
- (c) These both compounds are homologous series of aldehydes. These differ by $-CH_2$ group where first structure is Propanal and the 2nd structure is ethanal.

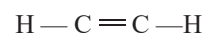
Propanal is the 3rd member of the homologous series of aldehyde and the ethanal is the 2nd member of the homologous series of aldehyde. **(1 Mark)**

4th member of this series will be butanal and its structure is: **(1 Mark)**



OR

- (a) Electron dot structure of ethyne is,



(1 Mark)

- (b) Difference between covalent and ionic compounds,
- (i) Physical nature: Ionic compounds are solids and are somewhat hard as compared to covalent compounds because of the strong force of attraction between the positive and negative ions. **(1 Mark)**

- (ii) Melting and boiling points: Ionic compounds have high melting and boiling points as compared to covalent compounds. **(1 Mark)**

10. (a) The gamete produced by human males is sperm. Sperm can either contain an X or a Y chromosome, which determines the sex of the offspring during fertilisation. While in the human females, egg, also known as ovum, are produced. **(1 Mark)**

- (b) No, a male child does not inherit an X chromosome from his father. This is because males have one X chromosome and one Y chromosome, while females have two X chromosomes. During fertilisation, the sperm from the father can either carry an X chromosome or a Y chromosome.

If the sperm carries an X chromosome, the resulting offspring will be female, while if it carries a Y chromosome, the resulting offspring will be male. Therefore, the X chromosome is inherited from the mother, not the father, in males. **(½ + 1 Marks)**

- (c) Females produce only one type of gamete, which is the egg cell or ovum having XX sex chromosomes.

(½ Mark)

11. (a) Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the temperature and other physical conditions remain constant. Mathematically, Ohm's Law can be represented as:

$$V = IR$$

where V is the potential difference applied across the ends of the conductor, I is the current flowing through the conductor, and R is the resistance of the conductor.

(1 Mark)

- (b) One ohm (1Ω) is defined as the resistance of a conductor in which a current of 1 ampere (1A) flows when a potential difference of 1 volt (1V) is applied across its ends.

(1 Mark)

- (c) Using Ohm's Law, we can find the resistance of the conductor as:

$$R = V/I$$

Given that the current flowing through the conductor is 0.5 A and the potential difference applied across its ends is 2 V.

Therefore, $R = 2 \text{ V}/0.5 \text{ A} = 4 \Omega$.

Therefore, the resistance of the conductor through which a current of 0.5 A flows when a potential difference of 2V is applied across its ends is 4 Ω.

(1 Mark)

12. (a) The resistance of a uniform cylindrical conductor of a given material depends on the following factors:

1. Length of the conductor: The resistance of a conductor is directly proportional to its length. This means that if the length of the conductor is increased, its resistance also increases. (½ Mark)
2. Cross-sectional area of the conductor: The resistance of a conductor is inversely proportional to its cross-sectional area. This means that if the cross-sectional area of the conductor is increased, its resistance decreases. (½ Mark)
3. Temperature of the conductor: The resistance of a conductor increases with an increase in temperature. This is because the increase in temperature causes an increase in the vibrations of the atoms in the conductor, which in turn increases the collisions between the free electrons and the atoms. (½ Mark)

4. Nature of the material: The resistance of a conductor depends on the nature of the material of which it is made. Some materials have a higher resistance than others. (½ Mark)

- (b) Resistivity, $\rho = 50 \times 10^{-8} \Omega \text{ m}$

$$R = \rho \frac{L}{A}$$

$$L = \frac{RA}{\rho}$$

$$= \frac{10 \times 3.14 \times (0.0001)^2}{50 \times 10^{-8}} = 0.628 \text{ m}$$

Hence, the length of the wire is 0.628 m.

(1 Mark)

OR

- (a) Electric power is the rate at which electrical energy is transferred or consumed by an electrical device. It is defined as the product of the voltage (V) applied across the device and the current (I) flowing through it. The SI unit of electric power is watt (W).

(1½ Marks)

- (b) $P = 2 \text{ kW}$, $t = 2 \text{ h}$

(i) $E = P \times t = 2 \times 2 = 4 \text{ kWh}$

(ii) $E = 2000 \text{ W} \times 2 \times 3600 \text{ s}$

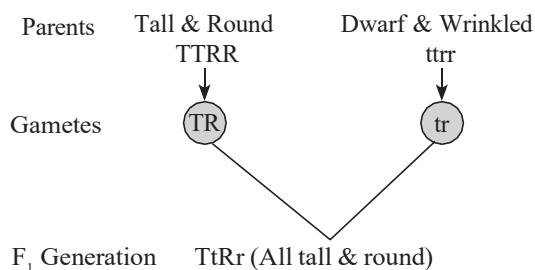
$$= 1.44 \times 10^7 \text{ J}$$

(1½ Marks)

13. (a) The natural water bodies have a balanced ecosystem with a variety of organisms that help to maintain water quality. While the aquariums are artificial environments that do not have producers and decomposers. Fish waste, uneaten food, and other organic matter can accumulate in the aquarium, leading to the build-up of harmful waste. This can be harmful to fish and other aquatic organisms living in the aquarium. Therefore, aquariums need to be cleaned regularly to remove this waste and maintain water quality. (1 Mark)

- (b) The ozone layer is getting depleted at higher levels of the atmosphere due to the release of chemicals, such as chlorofluorocarbons (CFCs). These chemicals react with ozone and break it down, leading to the depletion of the ozone layer. Harmful effects caused by the depletion of the ozone layer: An increase in the amount of harmful ultraviolet (UV) radiation reaching the Earth's surface leads to various health problems such as skin cancer, cataracts, and weakened immune systems. (2 Marks)

14. (a) The F₁ progeny of tall plants with round seeds and short plants with wrinkled seeds will all be tall with round seeds. This is because the dominant traits for both height and seed shape are expressed in the first generation. (1 Mark)



(b) The recessive traits in the above case are short height and wrinkled seeds. **(1 Mark)**

(c) The table shows the type of new combinations of plants obtained in the F₂ progeny, along with their ratio, if the F₁ generation is allowed to self-pollinate:

F₁ Generation TtRr (All tall & round)

$\downarrow$ Selfing

	TR	Tr	tR	tr
TR	TTRR Tall round	TTRr Tall round	TtRR Tall round	TtRr Tall round
Tr	TTRr Tall round	TTrr Tall wrinkled	TtRr Tall round	Ttrr Tall wrinkled
tR	TtRR Tall round	TtRr Tall round	ttRR Dwarf round	ttRr Dwarf round
tr	TtRr Tall round	Ttrr Tall wrinkled	ttRr Dwarf round	ttrr Dwarf wrinkled

Phenotypic ratio : Tall round (9) : Tall wrinkled (3) : Dwarf round (3) : Dwarf wrinkled (1) **(2 Marks)**

OR

If 1600 plants were obtained in the F₂ progeny, the number of plants having traits are as follows:

- (i) Tall with round seeds: $\frac{9}{16} \times 1600 = 900$ plants
 (ii) Short with wrinkled seeds: $\frac{1}{16} \times 1600 = 100$ plants
(1 + 1 = 2 Marks)

Conclusion: Mendel's experiment with pea plants showed that traits are inherited independently and that the dominant traits mask the expression of the recessive traits.

The ratio of the new combinations of plants obtained in the F₂ progeny is consistent with the laws of inheritance that Mendel proposed.

This experiment laid the foundation for modern genetics and provided insights into how traits are passed down from one generation to the next. **(2 Marks)**

15. (a) The rod gets displaced on passing current through it because of the interaction between the magnetic field produced by the current-carrying conductor and the external magnetic field. **(1 Mark)**

(b) The direction of the force on the conductor AB is determined by Fleming's Left Hand Rule. According to this rule, if we stretch the thumb, forefinger, and middle finger of our left hand mutually perpendicular to each other, then the forefinger represents the direction of magnetic field, the middle finger represents the direction of current, and the thumb represents the direction of the force on the conductor. **(1 Mark)**

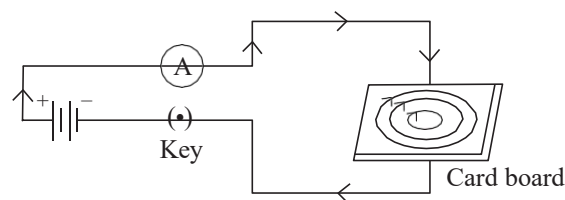
(c) (i) The rod will be displaced towards the east direction. **(1 Mark)**

(ii) Two devices that use current are:

1. Electric bulb - which uses electrical energy to produce light. **(½ Mark)**
2. Electric motor - which converts electrical energy into mechanical energy to perform work. **(½ Mark)**

OR

Maxwell's Right Hand Thumb Rule States



(2 Marks)

According to Maxwell's right hand thumb rule, if we hold the current carrying conductor in the right hand such that the thumb points in the direction of current, then the direction in which our fingers curl gives the direction of magnetic field. **(2 Marks)**